Assignment 1 – Internet Explorer

Client–Server Architecture and What Happens When You Open a Website

1. Client–Server Architecture

Client–Server Architecture is a network model where a client (user device) sends a request to a server, and the server processes the request and sends back a response. This model is widely used on the internet for websites and applications.

Main Components:

1. **Client:** The device used by the user such as a laptop, mobile phone, or tablet. Web browsers like Chrome, Edge, and Firefox act as clients.
2. **Server:** A powerful computer that stores websites, applications, and databases.
3. **Internet / Network:** The communication medium that connects clients and servers.
4. **Request and Response:** The client sends an HTTP request and the server returns an HTTP response.

Simple Client–Server Diagram:

```
   CLIENT (Browser)
        |
        |  HTTP Request
        v
   INTERNET / NETWORK
        |
        v
      SERVER
 (Website stored here)
        |
        |  HTTP Response
        v
   CLIENT (Web Page Displayed)
```

2. What Happens When You Open a Website

1. **Step 1: User enters URL** – The user types a website address such as https://www.google.com in the browser.
2. **Step 2: DNS Lookup** – The browser contacts the Domain Name System (DNS) to convert the domain name into an IP address.
3. **Step 3: Request Sent to Server** – The browser sends an HTTP/HTTPS request to the server.
4. **Step 4: Server Processes the Request** – The server retrieves website files such as HTML, CSS, JavaScript, and images.
5. **Step 5: Server Sends Response** – The server sends the webpage data back to the client.

6 **Step 6: Browser Displays Web Page** – The browser renders the page and displays it to the user.

Summary: Client–Server architecture allows communication between users and web servers. The browser sends requests, servers process them, and webpages are returned and displayed.